

The mine becomes the memory.

Spatial memory for emergency robots · IEEE RAS × IEEE AESS · TSYP14 · Phase 1 · Mines & tunnels

If the Writer is lost, the map is not.

In a mine without GPS or reliable connectivity, a robot's loss can erase its discoveries. LivingMap preserves those discoveries in radio beacons, so the next robot inherits useful knowledge and continues the mission.

01 / SYSTEM ARCHITECTURE

The mine keeps the memory.

One communication boundary. Three robot roles. Knowledge survives the Writer.

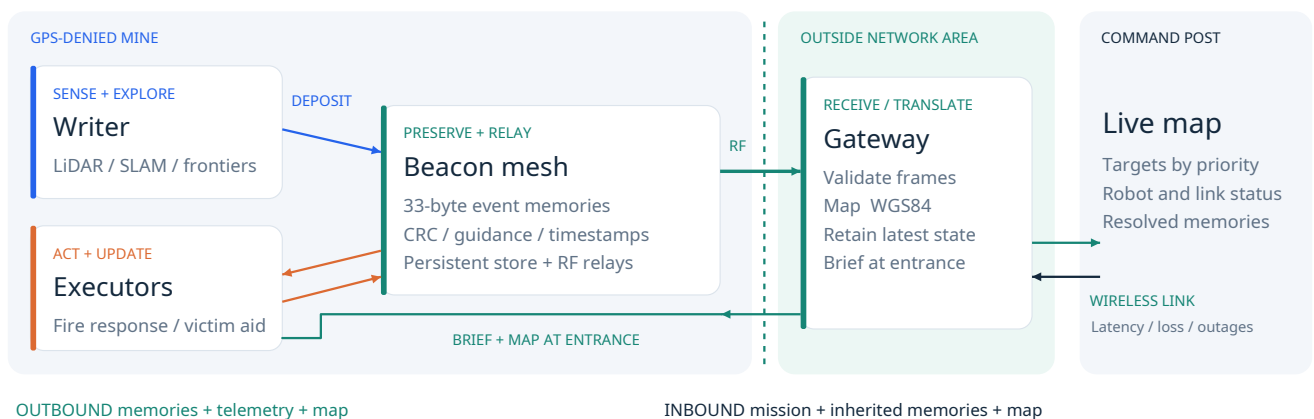


Figure 1. The Outside Network gateway is the bridge between the GPS-denied mine and the command post. System traffic crosses the simulated RF boundary; fault injection belongs to the separate harness.

What changes the mission

Design choice	Resulting behavior
Environment-owned memory	Beacon records persist independently of the Writer.
Safe-standoff guidance	Beacon coordinates mark a reached position; bearing and range locate the event.
WHO / WHAT / HOW	The command post chooses the robot and targets. Each Executor chooses its route.
Write-back into the beacon	Confirmed RESOLVED advertisements update later readers and the live map.

Prototype scope. Three simulated robots: Writer, fire-response Executor and rescue Executor. Ubuntu 24.04, ROS 2 Jazzy, Gazebo Harmonic, SLAM Toolbox and Nav2. Four event classes: victim, gas, fire and blocked path.

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Explore. Detect. Leave memory.

Writer autonomy · event sensing · deposition decisions · physical drop mechanism

The Writer builds a LiDAR occupancy map with SLAM Toolbox. Frontier exploration selects reachable boundaries between known free space and unknown cells, scores them by frontier size and travel distance, and sends goals to Nav2. Navigation remains inside the entrance boundary.

Writer outcome	Trigger	Response
HOME	No reachable frontier, or return reserve reached	Navigate to the entrance. A frontier-exhaustion return and a battery return do not establish the same coverage.
STUCK	Under 0.3 m progress in 45 s; or three failed home attempts	Report STUCK; retain radio and stored memories.
FAILED	Destruction or battery depletion	Transmit a 3 s dying gasp, then go silent.

Battery reserve: estimated charge needed to return $\times 1.5 + 10$ percentage points. This is a simulation policy, not measured battery endurance.

Deciding what to record

Configured zones simulate victim detection at 2 m, gas at 3 m, fire at 2.5 m and blocked paths at 1.4 m. The current default has two victims, two gas hazards, one fire and one blocked path. The automatic destroyed/stuck sequence waits for both victims, both gas hazards and fire before ending the sortie.

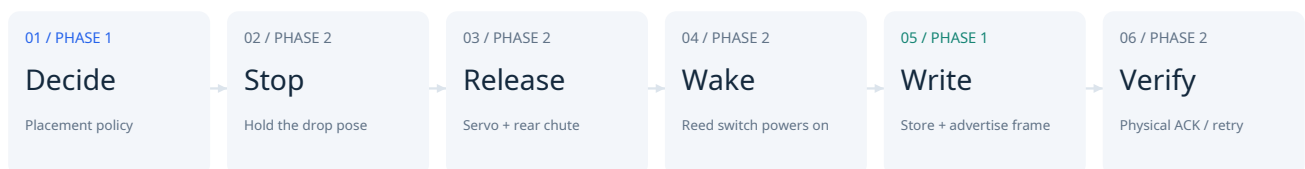
Priority	Placement condition	Memory role
Event	Sense an event; reuse a navigation beacon within 1 m when available	STANDOFF
1	First beacon inside the mine	ENTRANCE
2	Measured mesh uplink drops below 85%	RELAY
3	Three or more openings on a 1.6 m map ring; spacing permits	JUNCTION
4	No beacon within 6 m	BREADCRUMB

The Writer carries 30 beacons and reserves four for events. An event beacon records its own position, a guidance vector toward the event, severity, confidence, time, chain links and place type. Its position is a place the Writer has reached; this is not a proof of safety against unobserved hazards.

04 / BEACON DEPOSITION

Leave memory at a place already reached.

The beacon stays at a safe standoff. Its bearing and range locate the event.



30-beacon capacity · 4 reserved for events · event / entrance / relay / junction / breadcrumb policy

Figure 2. Phase 1 creates, writes and advertises beacon records at the Writer pose. The servo gate, rear chute, reed-switch wake and physical verification/retry are planned for Phase 2.

The hardware acceptance target is 20 drops without a jam. No physical dispenser has been demonstrated in Phase 1. Detailed design: docs/beacon_protocol.md; hardware milestones: docs/implementation_plan.md.

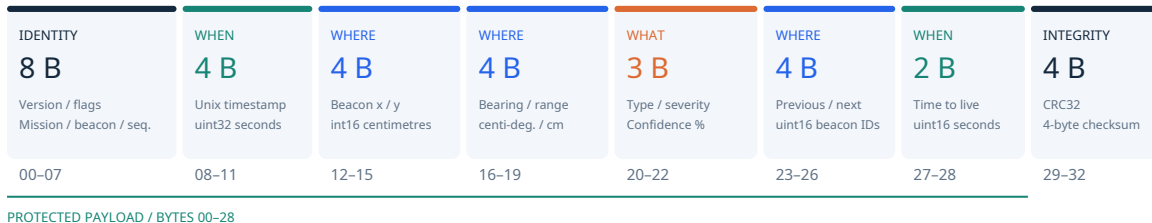
Compact memory. Resilient links.

Wire protocol v2 · modeled radio propagation · message aging

03 / BEACON MESSAGE

33 bytes that answer three questions.

Version 2 · network byte order · CRC32 over the first 29 bytes



Safe beacon position + guidance vector = event position. Previous leads out; next leads deeper in.

Figure 3. Field groups follow the implemented wire order. 33 bytes total; CRC32 protects bytes 00–28. Coordinates use centimetres; bearings use centi-degrees.

Read. STANDOFF guidance points from the beacon to its event. Navigation guidance points toward the next beacon; previous links lead out. Relays forward the same protected frame. The gateway rejects corrupt, duplicate or older frames.

Write. An Executor requests RESOLVED for the same memory identity. The beacon applies the flag to its own copy, preserving position and chain links, saves atomically and advertises the update. Hearing the updated advertisement confirms write-back. Requests retry every second for up to 20 s.

05 / RADIO MODEL

Walls shorten the link. Relays restore the path.

Simulation model from model.py · 8 dB per wall · no measured RF data claimed

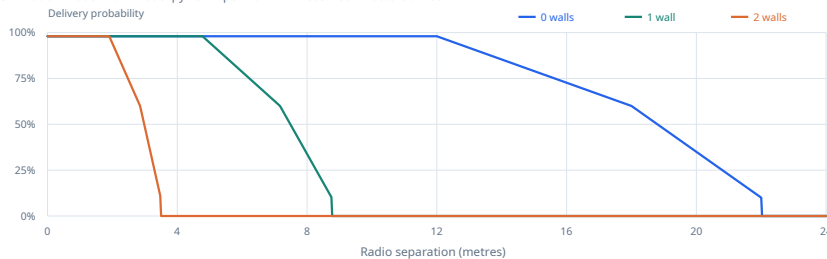


Figure 4. Model curves, not measured hardware data. Delivery is 98% through 12 m, 60% at 18 m, 10% at 22 m and zero strictly beyond the equivalent-distance limit.

Every crossed wall adds 8 dB, represented as an equivalent longer distance. Beacons advertise every 3 s and flood a new frame once per round. Telemetry uses the most reliable mesh route with three retries per hop. Recorded memories reach the gateway in 2–3 hops.

Memory	FRESH until	AGING until	Expires
Fire	1 min	5 min	15 min
Gas	2 min	10 min	30 min
Victim	5 min	15 min	60 min
Navigation	5 min	15 min	30 min
Blocked path	15 min	60 min	65,535 s

Expired memories are ignored; unresolved stale hazards are still avoided. Route costs penalize aging (+15%) and stale (+40%) memories. Live sensor contradiction of a memory is a Phase 2 design, not implemented behavior.

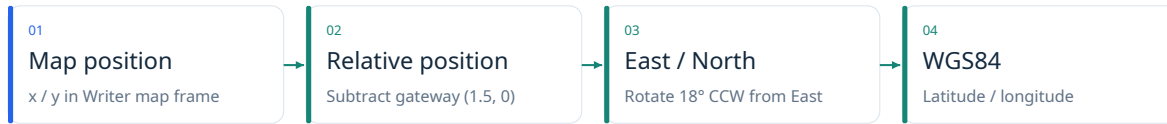
Translate outside. Act inside.

Gateway ownership · transactional briefing · beacon-guided mission execution

06 / FRAME TRANSLATION

A private map becomes a real-world location.

The gateway owns the GPS anchor and map heading; underground robots do not need GPS.



Example: victim (6.00, 2.20) m ENU (7.81, 0.23) m 36.8065020 N, 10.1815877 E

Figure 5. The gateway subtracts its map position, rotates by ENU heading and applies a local tangent-plane WGS84 conversion. Heading is counter-clockwise from East, not a compass bearing.

The GPS anchor is configured at 36.8065 N, 10.1815 E; gateway map position (−1.5, 0) m; map +X heading 18°. In the recorded example, victim (6.00, −2.20) m becomes ENU (7.81, 0.23) m and 36.8065020 N, 10.1815877 E. Numeric precision does not establish field accuracy: anchor error, heading error and SLAM drift remain.

The Outside Network Area

RECEIVE: validate and retain mesh frames, telemetry and the uploaded map. **TRANSLATE:** map coordinates to GPS and bearings to compass. **CARRY:** forward over a simulated 600 ms link with jitter, 2% loss, 32 KB/s capacity and outages. **BRIEF:** validate targets and deliver memories plus the latest map at the entrance.

02 / MISSION CONTINUITY

Explore. Preserve. Continue.

The command post chooses WHO and WHAT. Each robot decides HOW.

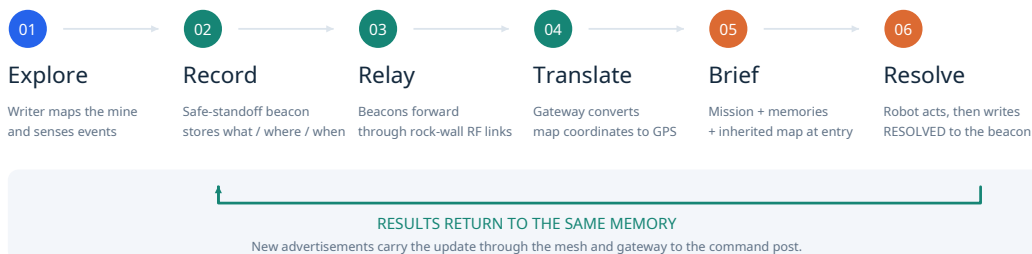


Figure 6. All system information passes through the gateway. Scenario placement and destruction/outage controls are explicitly simulation-harness functions.

Each Executor receives its own mission and a matching transactional seed of beacon frames and the Writer's last map before planning. Gateway briefing requires a ready robot in one-hop range. After entry, fresh radio advertisements update its local memory.

Executor policy and actions

Targets follow the command post's priority order: severity, confidence and freshness. The robot builds a beacon graph; driven links are preferred, inferred links cost 25% more, and map checks reject wall-crossing legs. Nav2 drives between waypoints. Gas, fire and blocked paths become keep-out discs of 2.0 m, 1.8 m and 0.7 m.

The fire robot acts for 5 s from standoff; the rescue robot assists for 3 s within its target tolerance. An action is followed by beacon write-back. Fire response is dispatched first; rescue follows the fire robot's return report or the controller's fallback conditions. Each robot attempts to return after processing its targets.

A completion state can include skipped targets or unconfirmed write-back, and return travel finishes later. Confirm success from target details, resolved memories and return reports together.

Evidence you can inspect.

Individual autonomous simulation runs · source-linked results · bounded claims

08 / SIMULATION EVIDENCE

Three handovers. The objectives continued.

4 October 2026 · one recorded run per scenario · elapsed to objective completion

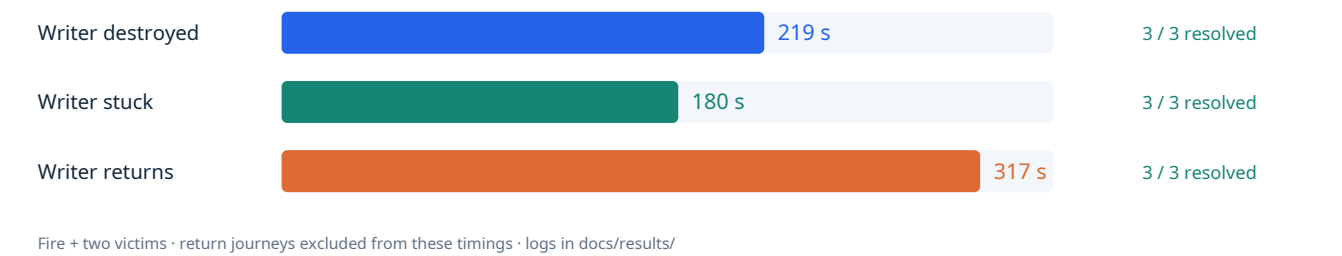


Figure 7. Watcher elapsed time to objective completion in three 4 October runs. All resolved one fire and two victims. These are single observations, not means or reliability estimates.

07 / RECORDED MISSION

One inherited map. Three distinct roles.

5 October 2026 · spacious layout / one gas hazard · final snapshot, retained trails only



SOURCE docs/results/run_spacious_state.json · all panels use recorded map coordinates (metres)

Figure 8. All panels use the same 5 October spacious-layout snapshot: recorded occupancy, beacon/event coordinates and retained trails. Final status shows three resolved event memories and both Executors back at the entrance.

Evidence	What is established	Boundary
4 October runs	Writer destroyed / stuck / home; fire and two victims resolved	One gas hazard; snapshot timing precedes completed returns.
5 October spacious snapshot	Writer lost; fire + two victims resolved; both Executors returned	One gas hazard; bounded retained trails, not full trajectories.
Relay destruction log	Five dependent beacon routes recovered	One recorded relay-removal scenario.
149 central tests; headless demo	Protocol, mesh, GPS, policies, persistence and information flow	No package-local ROS tests; not complete navigation coverage.

The current default includes two gas hazards. Historical recordings demonstrate their own layouts and do not certify every exploration order or the current configuration. A repository test pass and a MISSION COMPLETE banner are separate from confirmed end-to-end objectives.

Sources: docs/results/run_destroyed.log; run_stuck.log; run_return.log; run_spacious_state.json; beacon_destroyed.log. See docs/results/README.md for evidence scope. Reproduce with instructions.md and scripts/test.sh.

Handle failure. Build the hardware.

Failure response · current limitations · eight-week prototype plan

Failure	Implemented response	Evidence
Writer lost / stuck	Preserve independent beacon memory; dispatch with available knowledge	Recorded runs
Low battery / failed return	Reserve-based return; after retries, report STUCK	Central logic tests
Relay removed	Recompute RF routes; retained gateway copy remains available	Recorded log + tests
Outage	Keep newest state; forward on recovery; commands retry	Recorded run + tests
Corrupt / old frame	CRC rejection and sequence deduplication	Central tests
Expired / stale memory	Ignore expired; keep unresolved stale hazards out	Central tests
Target unreachable	Skip and report; continue remaining targets	Recorded behavior
Unconfirmed write-back	Retry, then report timeout and continue	Implemented behavior
Nav2 / simulator startup	Lifecycle activation and simulation-clock guards	Recorded diagnostics

Current limits

Zone sensing, ideal odometry, timed actions and modeled RF/link/battery behavior remain simulation abstractions. The system has one gateway, no gas-neutralization capability, and no live sensing check that contradicts old beacon memories. A battery-triggered return does not imply a complete map.

The current completion state can include partial target success. Beacon acknowledgement timeout does not prove a result was preserved; the command-post summary can overstate resolution. Sequential dispatch can proceed after a failed fire-robot return or timeout. These are explicit limitations, not validated safety guarantees.

Physical implementation

At least two physical robots are required: Writer and Executor. Planned bases use Raspberry Pi / ROS 2, motor controller, LiDAR, encoders and IMU. The Writer adds victim/gas/flame sensing and a servo dispenser. Battery radio beacons implement the same frame and flooding rules. The gateway adds a measured GPS anchor and cellular or Wi-Fi uplink. A third fire robot is optional.



Figure 9. Planned hardware milestones to 1 December 2026; none imply completed physical validation.

Two-robot planning estimate: 4,230 DT, approximately 4,650 DT with 10% contingency. Optional fire robot: +1,375 DT before contingency. Supplier quotations, permitted RF band and availability need validation. Acceptance includes valid real-radio frames, measured through-wall links, 20 jam-free drops and end-to-end failure trials.

Detailed documents: docs/failure_analysis.md; docs/implementation_plan.md; docs/architecture.md; docs/beacon_protocol.md; docs/coordinate_frames.md.